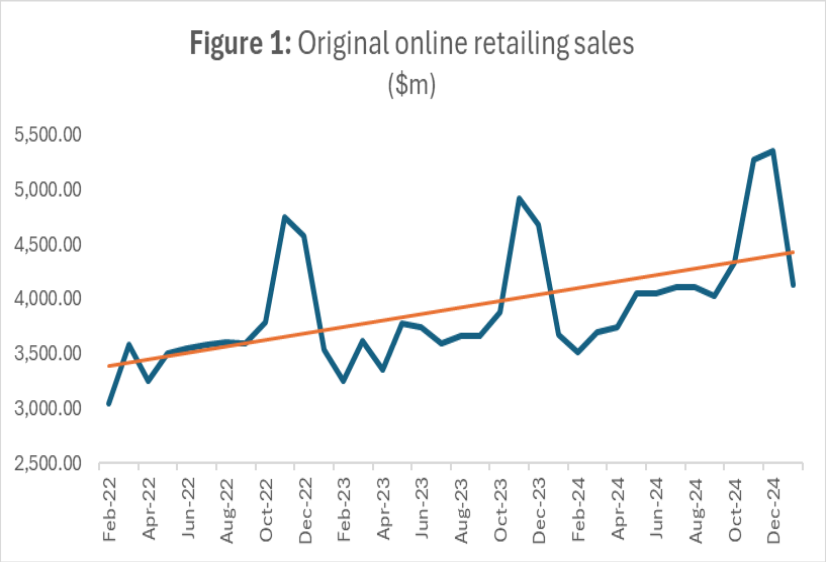
Report – Section B

1. Attribution

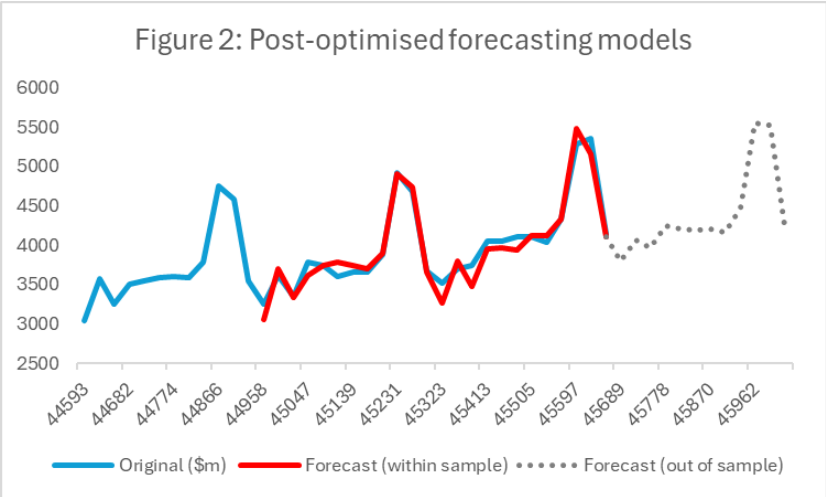
This report is designed to develop an optimal forecasting model for retail trade on behalf of the Australian Retailers Association. It utilizes data from [Retail Trade, Australia, February 2025 | Australian Bureau of Statistics] published by the Australian Bureau of Statistics, specifically drawing on Table 1. For Exercise 1, seasonally-adjusted figures are analyzed, while Exercise 2 focuses on original (non-adjusted) figures. All values are measured in millions of dollars, estimated monthly from February 2025 to January 2026.

2. SCOPE

To forecast online retailing sales in Australia, the Winters' Exponential Smoothing (WES) model was applied. This model is especially useful when the data shows both seasonal patterns and long-term growth or decline, which is exactly what we see in this case.



In Figure 1, sales tend to peak around December each year, likely due to holiday shopping, while the general level of sales gradually increases over time.



In *Figure 2*, the red line represents the fitted forecasts based on the post-optimized model, closely following the blue line (actual sales). The grey dotted line shows the projected values for future months (out-of-sample), maintaining the upward trend and capturing expected seasonal peaks. The model successfully predicts the recurring December spike and accounts for gradual increases over time.

3. Application

The Winters' method addresses this by using **three components**:

- a "level" component to capture the overall average of the series,
- a "trend" component to follow longer-term increases or decreases, and

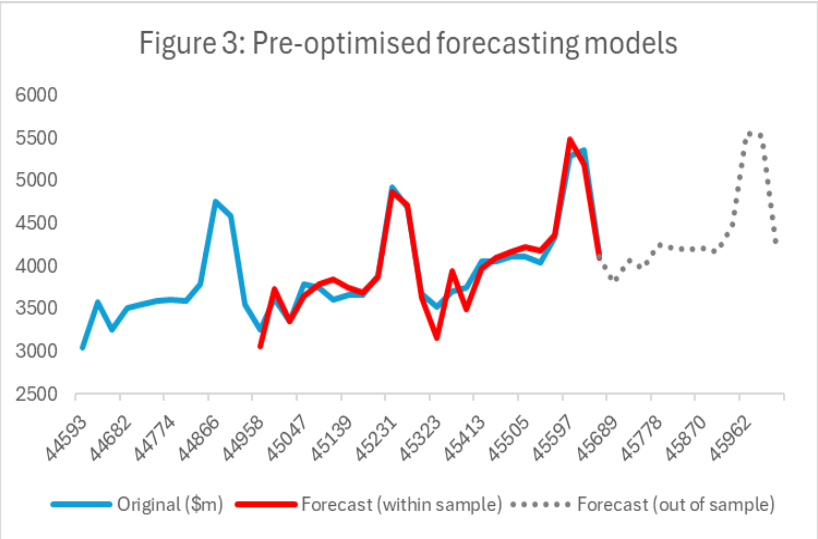
- a “seasonal” component that adjusts for repeating patterns throughout the year.

The **multiplicative** version of the WES method is applied, as the seasonal fluctuations in online retail sales vary proportionally with the sales level, rather than remaining constant over time. The formulas applied in the model are summarized below:

Component	**Formula**
Level (L_t)	$L_t = \alpha \times (Y_t \div S_{t-s}) + (1 - \alpha) \times (L_{t-1} + T_{t-1})$
Trend (T_t)	$T_t = \beta \times (L_t - L_{t-1}) + (1 - \beta) \times T_{t-1}$
Seasonality (S_t)	$S_t = \gamma \times (Y_t \div L_t) + (1 - \gamma) \times S_{t-s}$
Forecast ($\hat{Y}_{t+m}$)	$\hat{Y}_{t+m} = (L_t + m \times T_t) \times S_{t+m-s}$

Each component (level, trend, seasonality) is updated independently with its corresponding smoothing constant. To build the model, the initial parameters were optimized to minimize forecasting errors, improving model accuracy compared to the pre-optimized version.

To validate the model, calculation of mean squared error (MSE) was performed. This metric evaluates forecasting accuracy, with MSE penalizing larger errors more heavily due to squaring. By optimizing the model parameters, the MSE was reduced, leading to a forecast that more accurately captures the underlying sales patterns and seasonal peaks.



4. Analysis

Figure 3 shows the performance of the multiplicative Winters’ Exponential Smoothing (WES) model with pre-optimised parameters ($\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$). While the model captures the seasonality and upward trend in turnover, it struggles to accurately predict peak periods, particularly during later months. Forecasts tend to underestimate during high-demand months and overestimate during slower periods, highlighting the need for further model refinement.

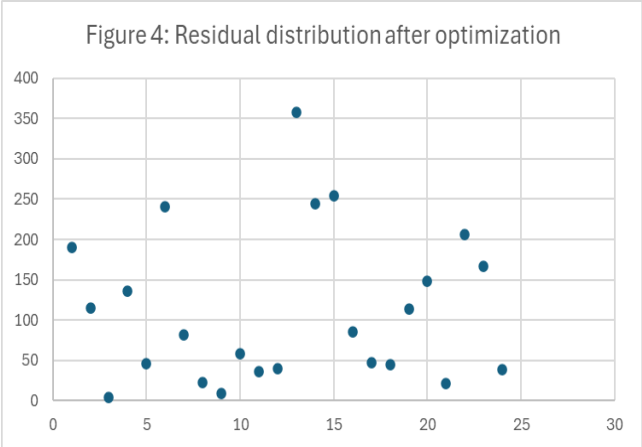
Comparatively, Figure 2 presents the post-optimized model where the smoothing parameters have been adjusted to minimize forecast error. This model shows a much closer alignment between the forecasted and actual data across both the within-sample and out-of-sample periods. Notably, the post-optimized forecast captures the major seasonal peak more accurately, especially around the significant turnover increase observed at the end of the sample.

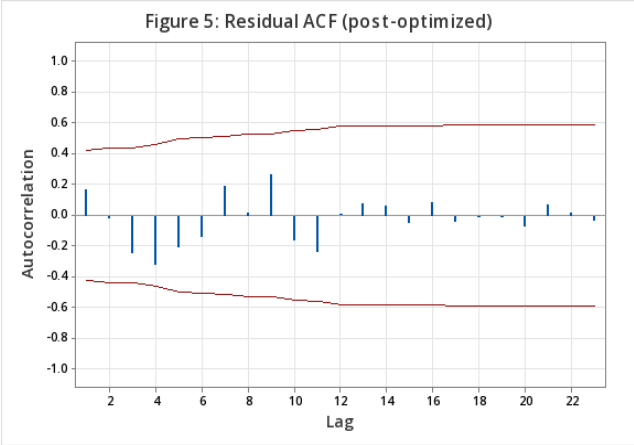
WES	*MAE*	*MSE*	*MAPE*
Pre-optimized	112.9435	21388.84	2.903561

Post-optimized	99.85112	16706.77	2.540448

The improvement is confirmed by validation statistics, with MSE decreasing from 21,388.84 to 16,706.77, alongside lower MAE and MAPE. This indicates that the post- optimized model reduces forecasting errors and better fits the data, enhancing its reliability for business forecasting. In the out-of-sample period, the forecast maintains a steady upward trend with clear seasonality, supporting the model's predictive strength for future use.

5. Articulations of Issues





Most residuals in Figure 4 are scattered above zero, with several points exceeding 300, suggesting the presence of outliers. The ACF plot in Figure 5 shows significant autocorrelations at multiple lags, indicating that some time-based patterns remain uncaptured. While these issues suggest minor model limitations, the overall post-optimized model remains sufficiently reliable for forecasting purposes.

6. Critique

The optimized Winters' Exponential Smoothing (WES) model aimed to capture both the trend and seasonal components of the retail turnover data. While the trend component was relatively weak, the model effectively accounted for recurring seasonal patterns, leading to strong predictive accuracy overall.

Model	MAE	MSE
WES (post-optimised)	99.85	16,706.77
HES (post-optimised)	372.05	276,173.59

Comparing WES with Holt's Exponential Smoothing (HES) reinforces the importance of seasonality in the dataset. WES substantially outperformed HES, achieving much lower MAE and MSE values. This indicates that seasonality remains a critical driver of retail turnover patterns, and models that explicitly capture this component deliver superior forecasting results.

Both models assume stable historical patterns, which external factors like inflation and shifting consumer behavior can disrupt. To stay relevant, WES forecasts should be regularly adjusted with expert judgment and updated economic insights to better support business decisions.

7. Position

The optimized WES model demonstrates superior forecasting accuracy compared to HES, effectively capturing the seasonal patterns critical to retail turnover. Its lower MAE and MSE values confirm its suitability for short-term business forecasting. However, the model's reliance on historical patterns must be carefully managed in dynamic environments. External factors such as inflation and changing consumer behaviors may distort seasonal effects over time. To ensure forecasts remain strategically valuable, WES outputs should be regularly reviewed and supplemented with judgmental adjustments based on current market insights.